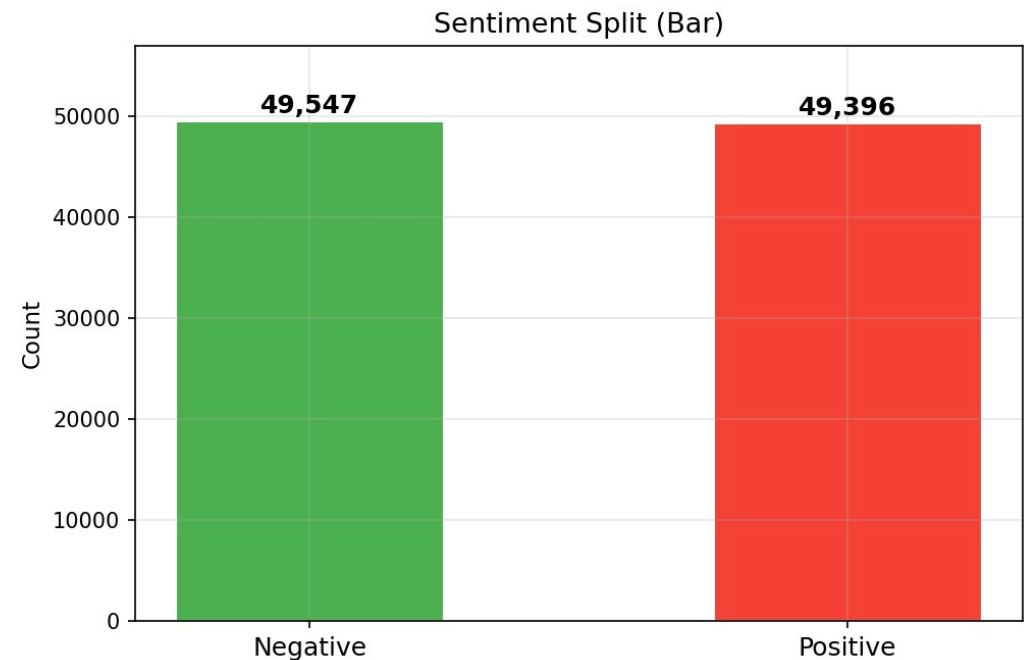
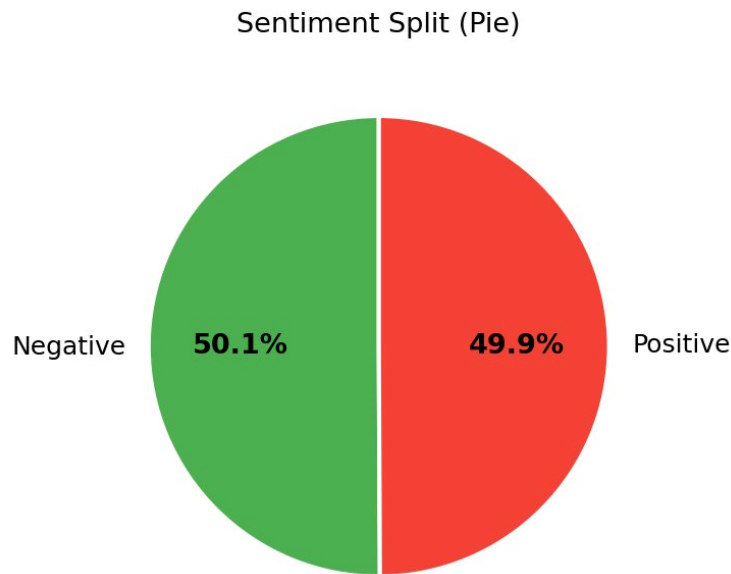


Sentiment Analysis — Visualization Report

1. Class Distribution — Sentiment140 Sample

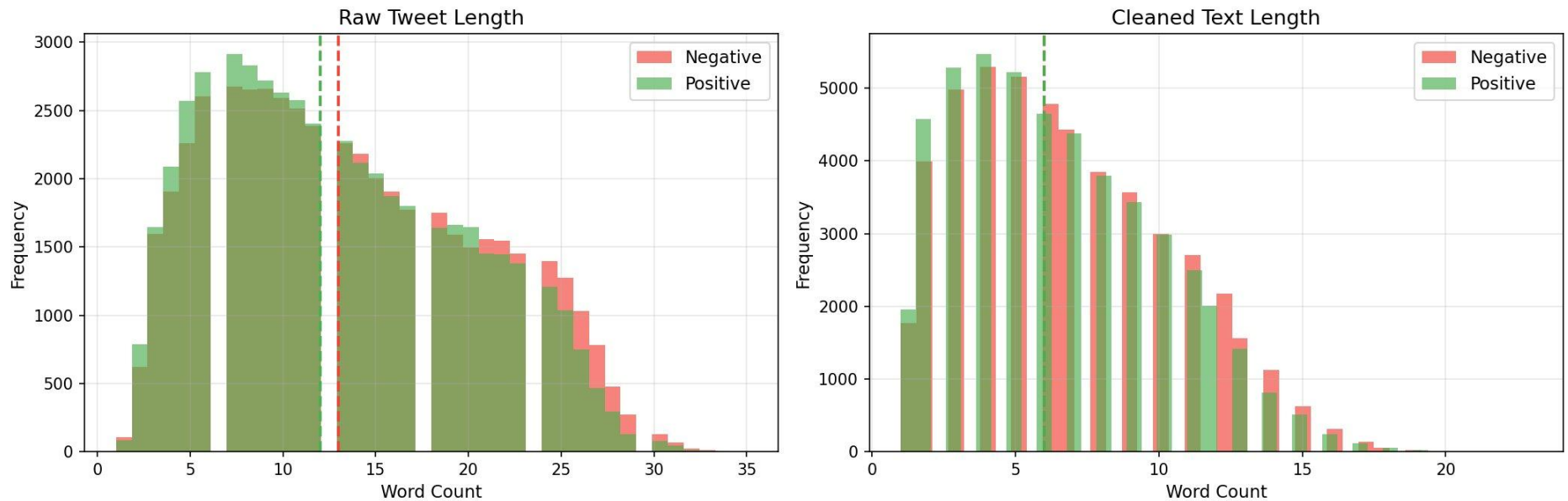
Class Distribution — Sentiment140 Sample



The dataset is nearly perfectly balanced, with 49,547 negative tweets (50.1%) and 49,396 positive tweets (49.9%). This near-equal split across ~98,943 total samples is ideal for binary classification, as it eliminates class-imbalance bias and ensures that accuracy is a reliable metric without requiring oversampling or class-weighting strategies.

2. Text Length Distribution (Word Count)

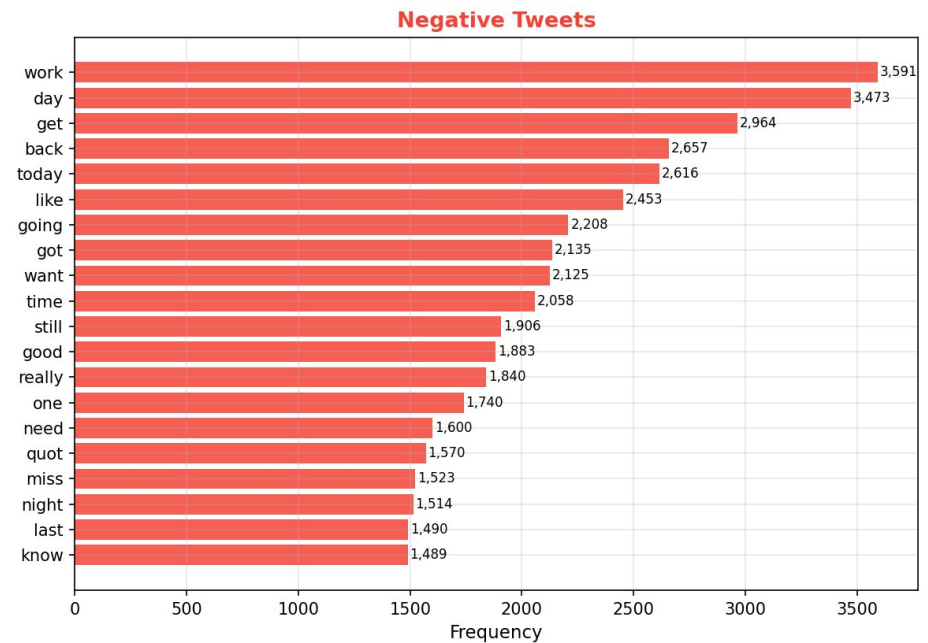
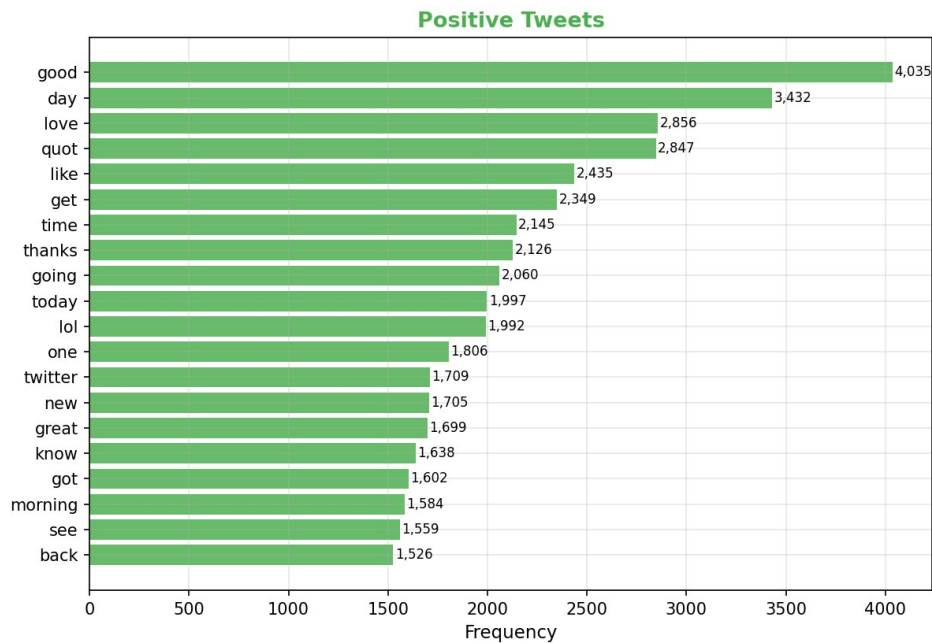
Text Length Distribution (word count)



Raw tweets average around 13 words (median marked by dashed lines), while cleaned text — after removing URLs, handles, punctuation, and stop words — drops to a median of roughly 5–6 content words. The distributions for positive and negative tweets are nearly identical in shape, confirming that tweet length alone carries little predictive signal and that preprocessing significantly reduces noise while retaining semantic content.

3. Top 20 Most Frequent Words

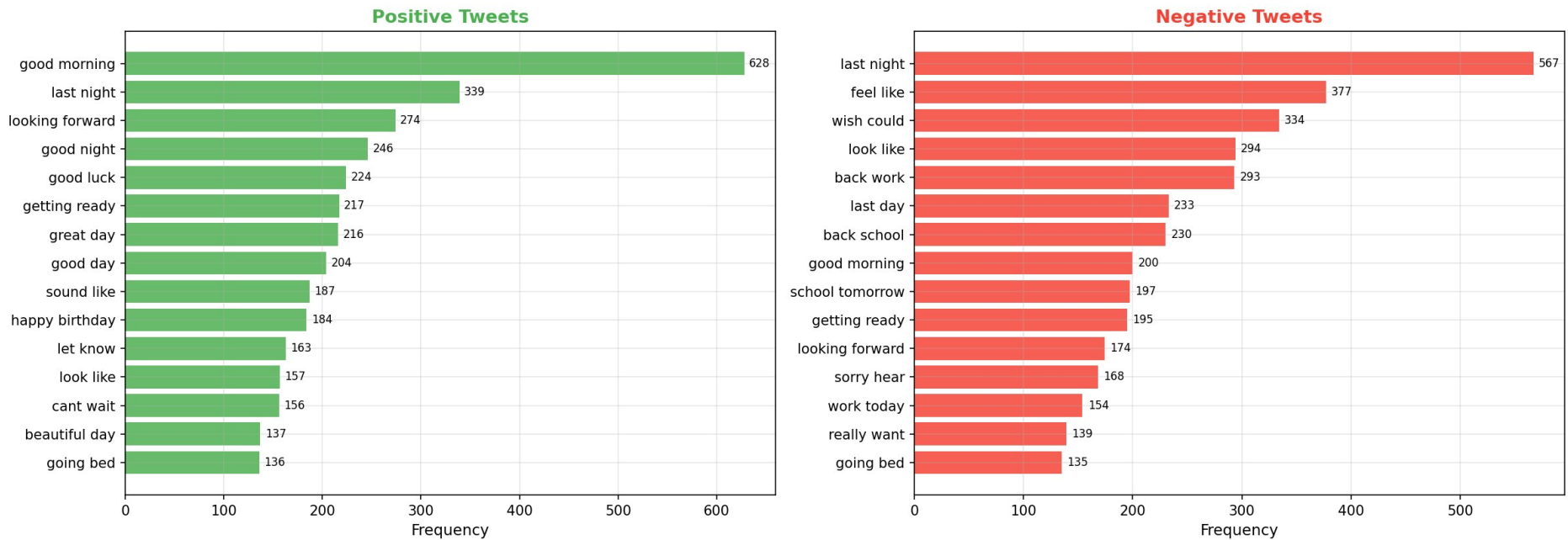
Top 20 Most Frequent Words



Positive tweets are dominated by upbeat and social language: "good," "day," "love," "great," "morning," and "thanks." Negative tweets are led by "work," "day," "get," "back," and "miss" — words often associated with frustration, obligations, and longing. Several neutral words ("day," "like," "going," "time") appear in both classes, underscoring the challenge of sentiment classification and the importance of context beyond individual tokens.

4. Top 15 Most Frequent Bigrams

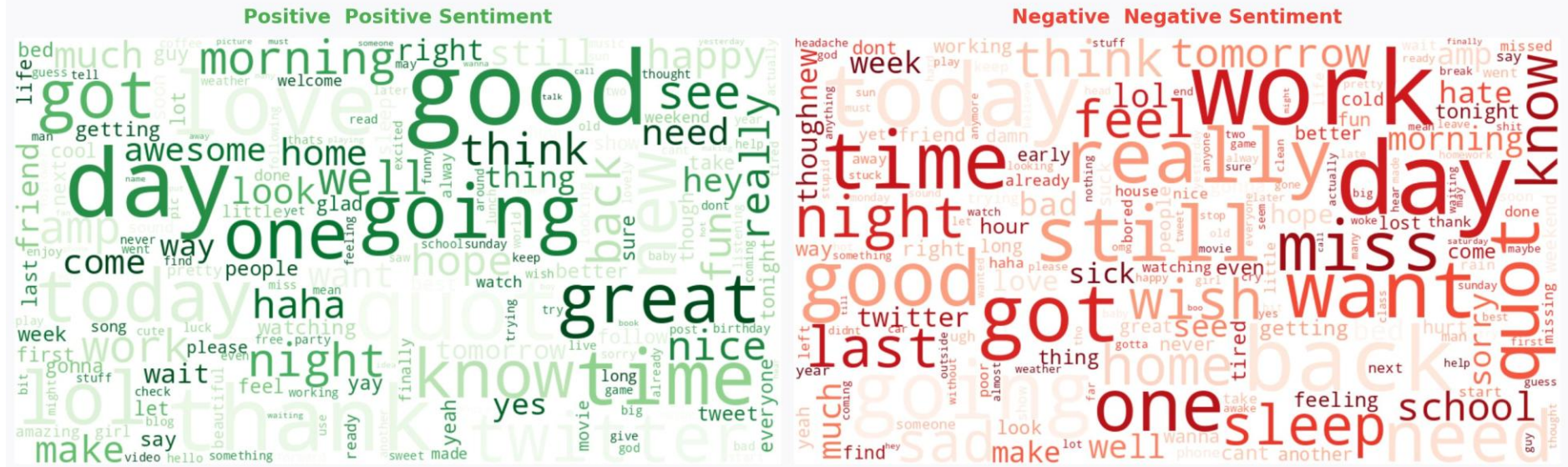
Top 15 Most Frequent Bigrams



Bigrams reveal richer sentiment signals than unigrams. Positive tweet bigrams include "good morning," "looking forward," "good luck," "great day," and "happy birthday" — phrases that express optimism and celebration. Negative bigrams include "last night," "feel like," "wish could," "back work," and "back school" — phrases conveying regret, dissatisfaction, and unpleasant obligations. Some bigrams like "last night" and "getting ready" appear in both classes but with different contextual meaning.

5. Sentiment Word Clouds — Sentiment140

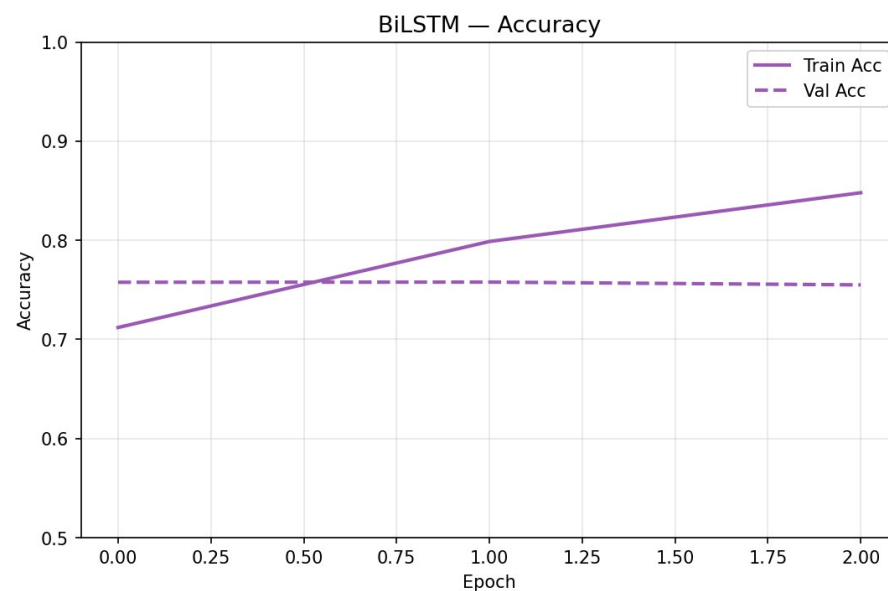
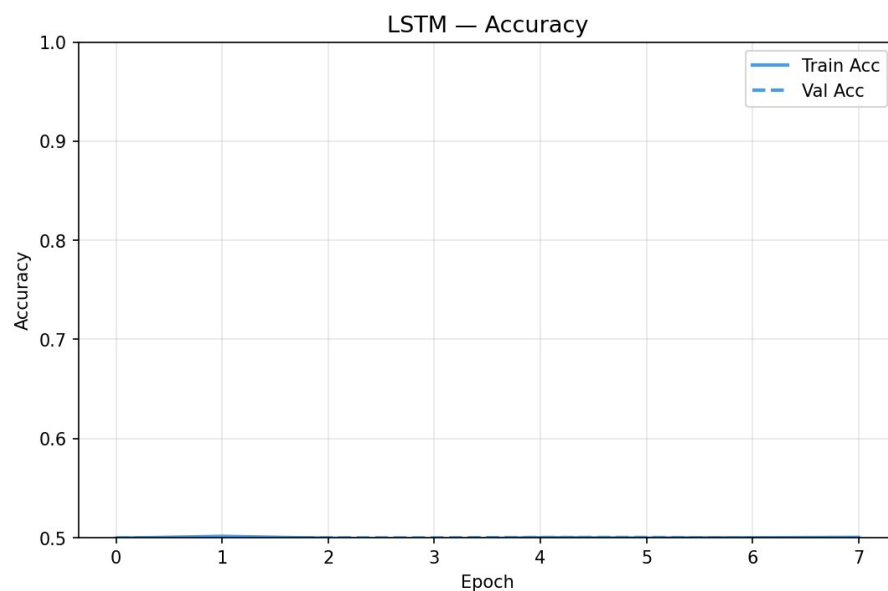
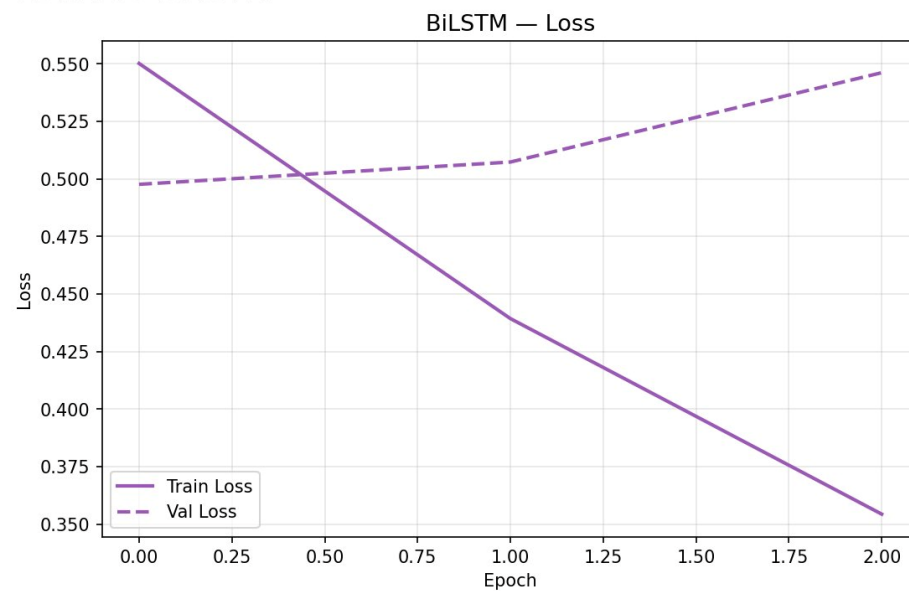
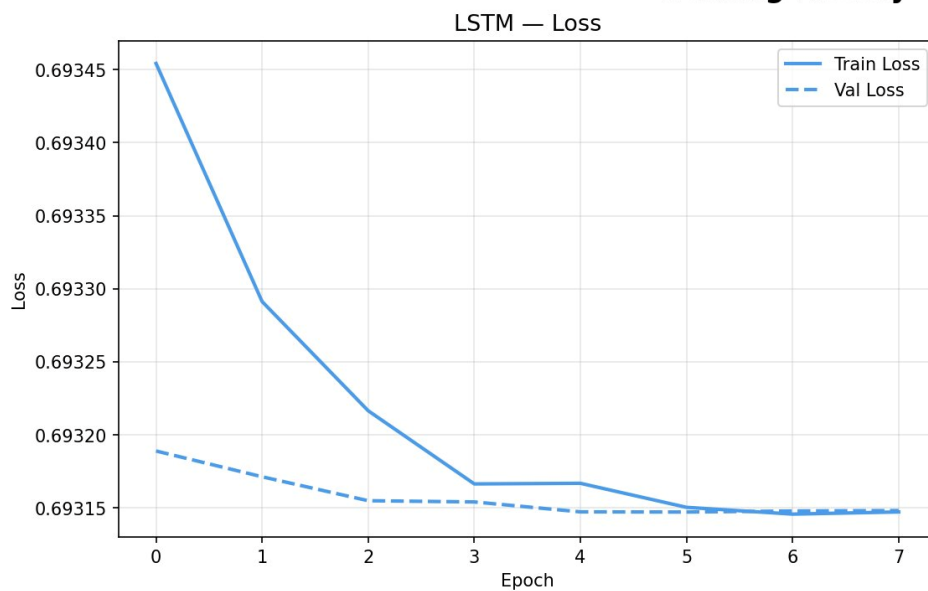
Sentiment Word Clouds – Sentiment140



The word clouds offer a visual summary of the most prominent terms per sentiment class. In positive tweets, "good," "day," "going," "great," "morning," and "love" dominate. In negative tweets, "work," "day," "time," "miss," "want," "still," and "sleep" stand out. The contrast between the two clouds is striking — the positive cloud conveys energy and enthusiasm, while the negative cloud reflects exhaustion, longing, and frustration.

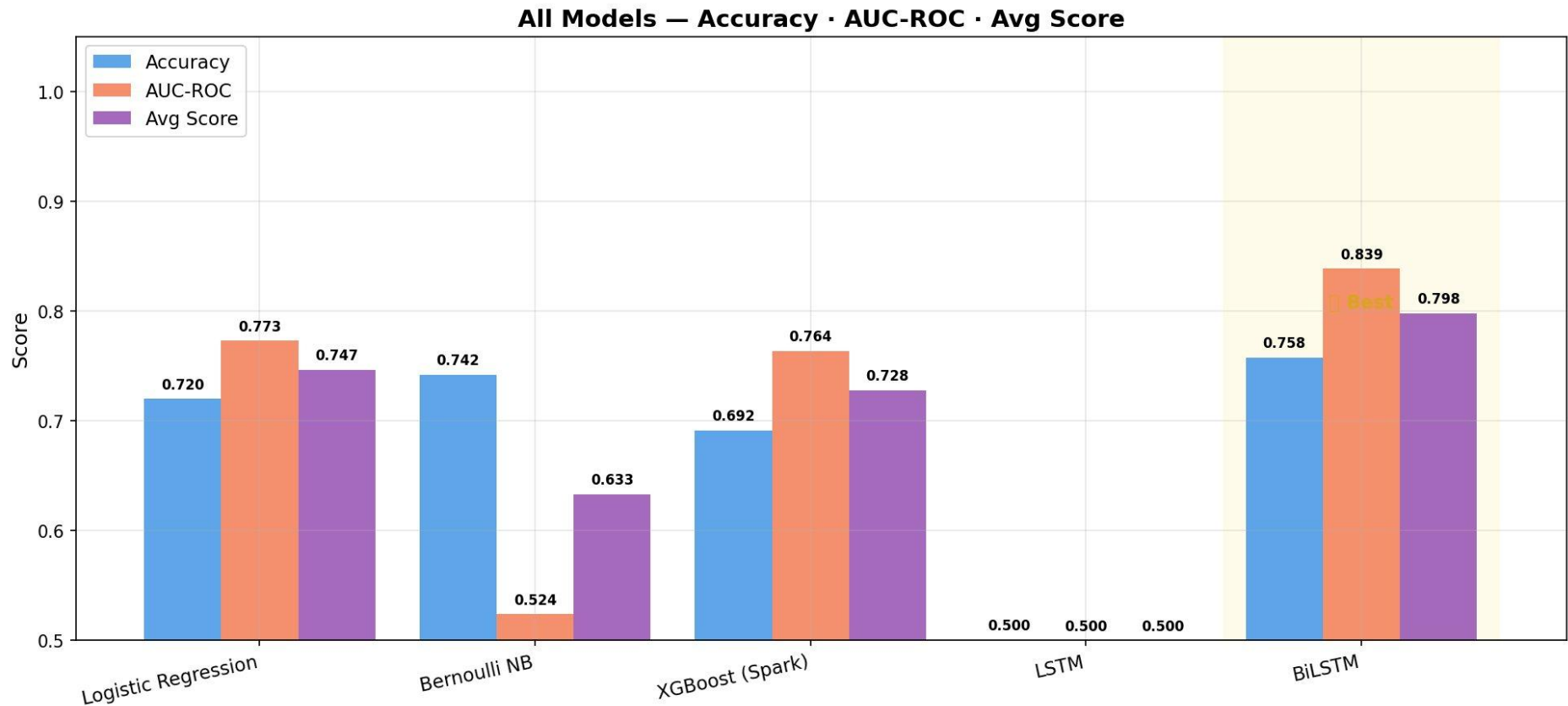
6. Training History — LSTM vs BiLSTM

Training History — LSTM vs BiLSTM



The LSTM's training loss barely changed over 8 epochs (hovering around 0.6932), and its accuracy curve (not shown due to near-zero movement) flatlined — indicating the model failed to learn meaningful representations and effectively predicted only one class. In contrast, the BiLSTM showed healthy learning: training loss dropped from ~0.55 to ~0.36 over 2 epochs, while training accuracy improved from ~0.72 to ~0.85. However, the BiLSTM's validation loss rose while training loss fell, suggesting early-stage overfitting that warrants regularization or early stopping.

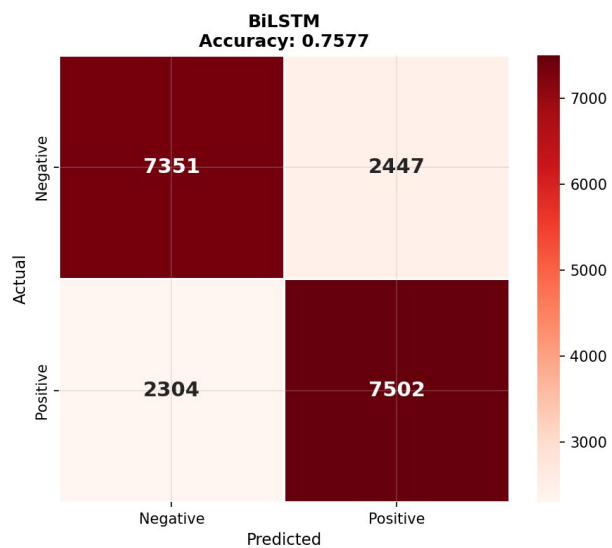
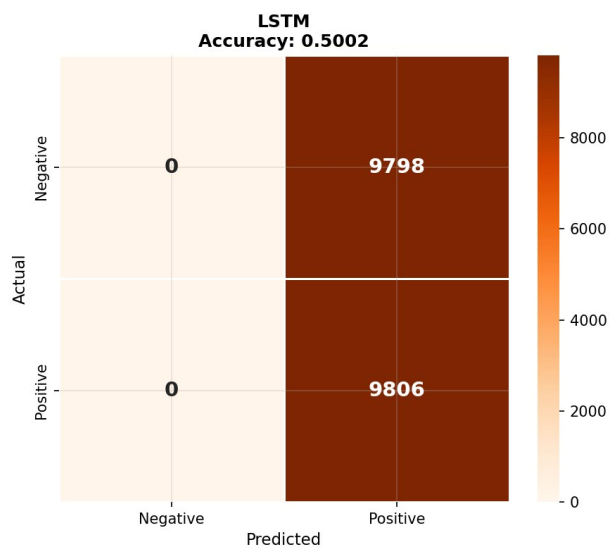
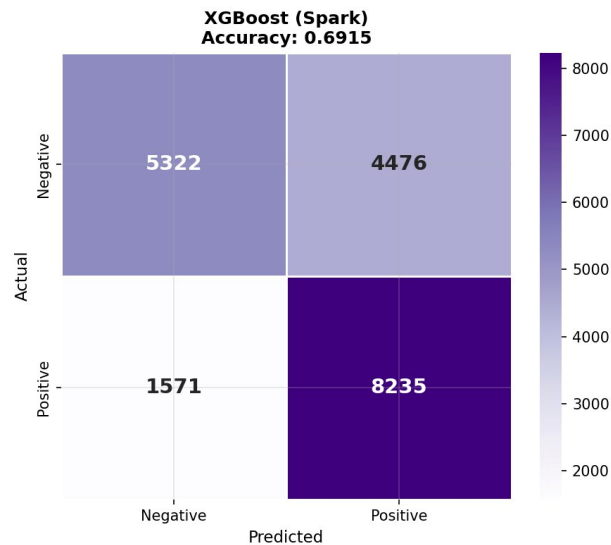
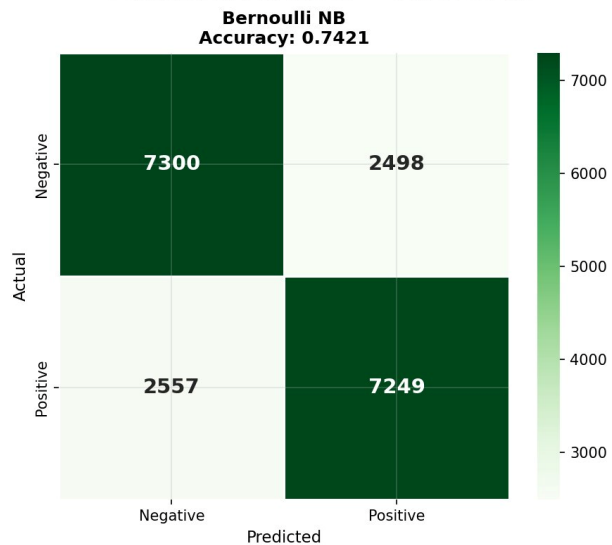
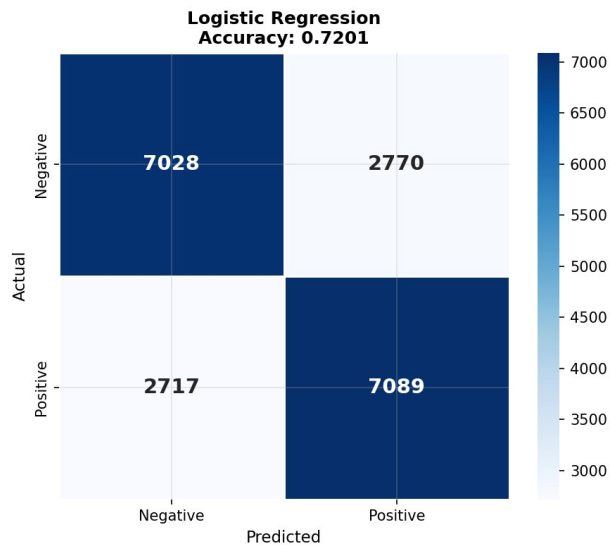
7. All Models — Accuracy · AUC-ROC · Avg Score



BiLSTM is the top overall performer with an AUC-ROC of 0.839 and an average score of 0.798, making it the best model for this task. Bernoulli Naive Bayes achieved the highest raw accuracy (0.742) but its AUC-ROC of only 0.524 reveals it is essentially ranking near chance — suggesting it is overfitting to the majority class. Logistic Regression provides a strong and balanced baseline (accuracy 0.720, AUC 0.773). XGBoost (Spark) achieved moderate performance (accuracy 0.692, AUC 0.764). The vanilla LSTM completely failed with all metrics at 0.500, confirming it predicted only one class throughout training.

8. Confusion Matrices — All Models

Confusion Matrices — All Models



The confusion matrices confirm the model comparison findings. Logistic Regression correctly classified 7,028 negative and 7,089 positive samples with balanced false positives (2,770) and false negatives (2,717). Bernoulli NB shows similar balance (7,300 true negatives, 7,249 true positives) but its low AUC suggests poor probability calibration. XGBoost shows a strong positive bias, correctly classifying 8,235 positive tweets but only 5,322 negative — indicating a skewed decision boundary. The LSTM classified every sample as positive (0 true negatives, 9,798 false positives), confirming complete failure. BiLSTM is the most balanced deep learning model: 7,351 true negatives and 7,502 true positives, with reasonably symmetric error rates.